

A rectangle interpretation of Vieta jumping for IMO 1988 Problem 6

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Expository note

Abstract

The standard Vieta-jumping solution of IMO 1988 Problem 6 replaces an ordered positive integer pair (a, b) by $(ka - b, a)$, where $a^2 + b^2 = k(ab + 1)$. This note interprets $ka - b$ as the width remaining after a b -square is removed from a strip of k adjacent $a \times b$ rectangles. We give the integer and descent arguments that justify repeating this construction until one coordinate is zero, proving that k is a perfect square. Greatest-common-divisor preservation also yields the known identity $k = \gcd(a, b)^2$. The purpose is expository: to give an area interpretation of the classical descent. Historical priority for this presentation is not claimed.

1 Introduction

Problem 6 of the 1988 International Mathematical Olympiad asks for the following result [1].

Theorem 1 (IMO 1988, Problem 6). *Let a, b be positive integers such that $ab + 1$ divides $a^2 + b^2$. Then*

$$k = \frac{a^2 + b^2}{ab + 1}$$

is a perfect square.

Classical solutions use an integer descent, commonly called Vieta jumping; Campbell's 1988 treatment also proves a greatest-common-divisor strengthening [2]. The construction below represents the same descent with rectangle areas. Its geometric role is to make the replacement length visible; integrality, preservation of the equation, and termination are justified explicitly. Section 4 discusses the relation to earlier presentations.

2 The rectangle construction and descent

By symmetry, suppose $1 \leq a \leq b$. The quotient k is a positive integer, and the defining equation is

$$a^2 + b^2 = kab + k. \tag{1}$$

Thus the two squares have the same total area as k copies of an $a \times b$ rectangle together with k unit squares. Place the rectangles side by side, each with width a and height b , to form a strip of width ka . Set

$$c = ka - b. \tag{2}$$

The following lemma ensures that a $b \times b$ square fits in the strip and that the resulting width gives a smaller admissible pair.

Lemma 2. *If $1 \leq a \leq b$ and k are integers satisfying (1), then $c = ka - b$ is an integer with $0 \leq c < a$, and*

$$c^2 + a^2 = k(ca + 1). \quad (3)$$

Proof. Integrality follows from (2). Suppose $c < 0$. Then $b - ka \geq 1$, so (1) gives

$$k = a^2 + b(b - ka) \geq a^2 + b > b.$$

But $b \geq ka + 1 \geq k + 1$, because $a \geq 1$ and $k \geq 1$. This is a contradiction, so $c \geq 0$.

Removing the b -square from the strip now leaves a $b \times c$ rectangle, as in Figure 1. Its area satisfies

$$bc = b(ka - b) = kab - b^2 = a^2 - k, \quad a^2 = bc + k. \quad (4)$$

Using $b + c = ka$, we obtain

$$a^2 + c^2 = bc + k + c^2 = c(b + c) + k = kac + k = k(ac + 1),$$

which proves (3). Finally, since $b > 0$ and $k > 0$,

$$0 \leq c = \frac{a^2 - k}{b} < \frac{a^2}{b} \leq a. \quad (5)$$

□

Proof of Theorem 1. Apply Lemma 2 to replace (a, b) by $(c, a) = (ka - b, a)$. The equation and the value of k are preserved, and $c < a \leq b$ implies

$$c + a < a + b.$$

If $c > 0$, the new pair is again ordered, positive, and integral, so the lemma applies again. An infinite repetition would give a strictly decreasing sequence of positive integer sums, which is impossible. Since every positive pair admits a next move, a move must eventually produce a zero coordinate.

Write the terminal pair as $(0, x)$, where x is a positive integer. The preserved equation becomes

$$0^2 + x^2 = k(0 \cdot x + 1),$$

so $k = x^2$. The zero coordinate is an auxiliary endpoint of the descent; the initial pair is positive as required. The case $k = 1$ is included: $(1, 1) \mapsto (0, 1)$. □

Corollary 3 (Known strengthening; see Campbell [2]). *Under the hypotheses of Theorem 1,*

$$k = \gcd(a, b)^2.$$

Proof. Each move preserves the greatest common divisor, since $\gcd(ka - b, a) = \gcd(a, b)$. At the terminal pair $(0, x)$ it equals x . Thus $x = \gcd(a, b)$ and $k = x^2 = \gcd(a, b)^2$. □

3 A worked example

For $(a, b) = (30, 112)$, the quotient is $13444/3361 = 4$. The complete descent is

$$(30, 112) \mapsto (8, 30) \mapsto (2, 8) \mapsto (0, 2).$$

This numerical orbit also appears in Wu's hyperbola visualization [3]. Table 1 records the corresponding area identities. The number of rectangles remains four while the integer lengths decrease.

At the final construction, $ka = b$, so the strip coincides with the b -square. No positive-width rectangle remains, and (4) reduces to $a^2 = k$. Figure 2 illustrates this endpoint. The diagrams use proportional dimensions within each construction; different constructions may use different scales.

A new side appears in the leftover strip

Example: $a = 8$, $b = 30$, $k = 4$. Dashed lines mark the original rectangles.

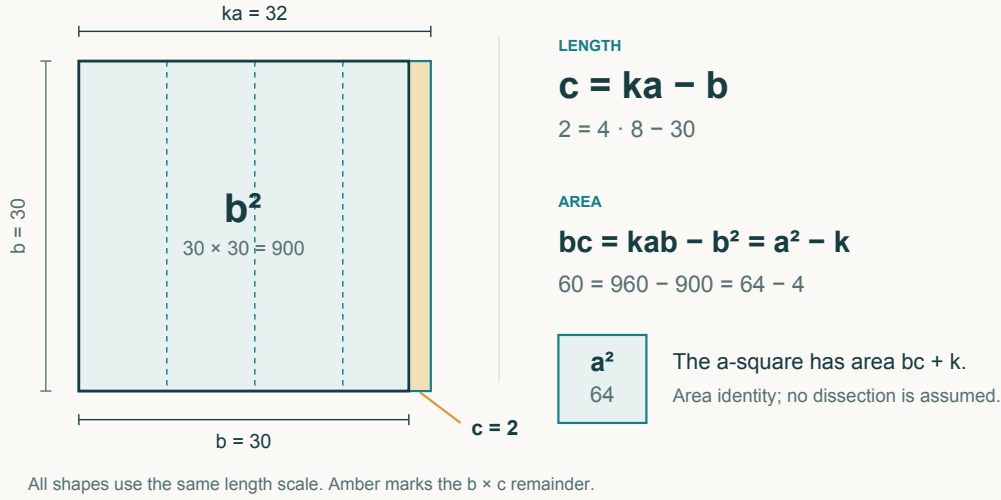


Figure 1: The strip construction for $(a, b, k) = (8, 30, 4)$. Four rectangles form a strip of width 32; removing the 30-square leaves width 2. The shapes have a common length scale. Equation (4) compares areas; no dissection of the smaller square is asserted.

Table 1: The descent for $k = 4$.

Current pair	$c = 4a - b$	$a^2 = bc + 4$	Next pair
(30, 112)	8	$900 = 896 + 4$	(8, 30)
(8, 30)	2	$64 = 60 + 4$	(2, 8)
(2, 8)	0	$4 = 0 + 4$	(0, 2)

At the boundary, the leftover width vanishes

Terminal jump: $(a, b) = (2, 8)$ becomes $(0, 2)$. Here $ka = b$, so $c = 0$.

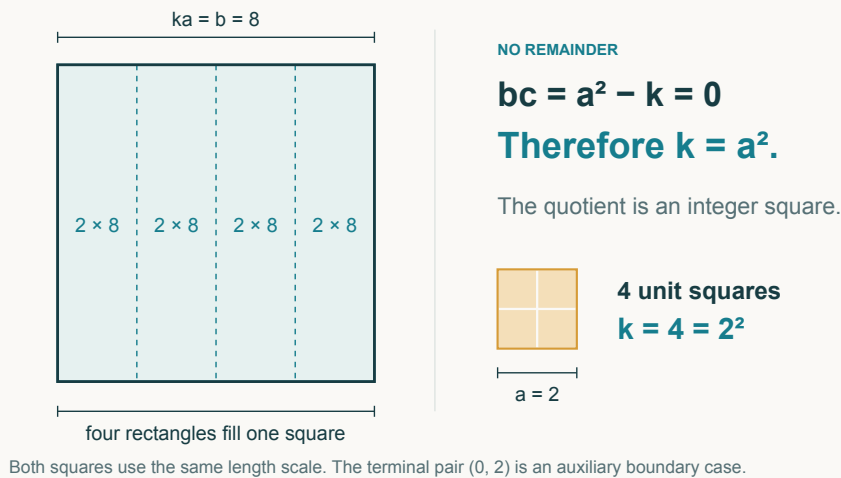


Figure 2: The final jump $(2, 8) \mapsto (0, 2)$. Four 2×8 rectangles fill an 8-square. The accompanying 2-square, drawn at the same scale, has area $k = 4$.

4 Relation to earlier work

Regarding (1) as a quadratic in its second coordinate gives

$$t^2 - kat + (a^2 - k) = 0.$$

Its roots are b and $ka - b$. Exchanging these roots is the classical Vieta jump. The strip construction interprets the second root as a remaining width, while the arithmetic transformation is unchanged.

Campbell's early treatment includes the greatest-common-divisor conclusion [2]. Wu visualizes Vieta jumping through integer points on a hyperbola [3]. A 2016 online discussion by Moody and contributors gives a remainder formulation of the same descent and includes a separate geometric attempt [4]; that discussion is background, not a premise of the proof here.

The contribution of this note is the repeated-rectangle exposition of that established descent. The references provide context rather than an exhaustive priority investigation. They do not establish that this particular area presentation is historically new, and no claim of priority is made. The argument uses both geometry and integer arithmetic; the diagrams alone do not prove the theorem.

Supplementary material

LaTeX source, SVG diagrams and matching vector PDF exports, diagram-generation scripts, and finite arithmetic checks are available in the accompanying repository.¹ The README provides a visual exposition with additional diagrams. The arithmetic script verifies the displayed examples and all admissible pairs with $1 \leq a \leq b \leq 400$. These finite checks can detect transcription errors; the general result follows from the proof above.

References

- [1] International Mathematical Olympiad. *1988 problems*, English, Problem 6, 1988. Official problem paper.
- [2] John Campbell. A solution to 1988 IMO question 6. *Mathematics Competitions* **1**(2) (1988), 29–32. Original article.
- [3] Kyle Wu. Vieta jumping: visualization and intuition. *Parabola* **60**(1) (2024), 1–7. Publisher's PDF.
- [4] Rutger Moody and contributors. Math Olympiad 1988 problem 6, canonical solution 2 without Vieta jumping. *Mathematics Stack Exchange*, 29 August 2016. Question and discussion.

¹<https://github.com/pranman/imo-1988-6-geometric-proof>